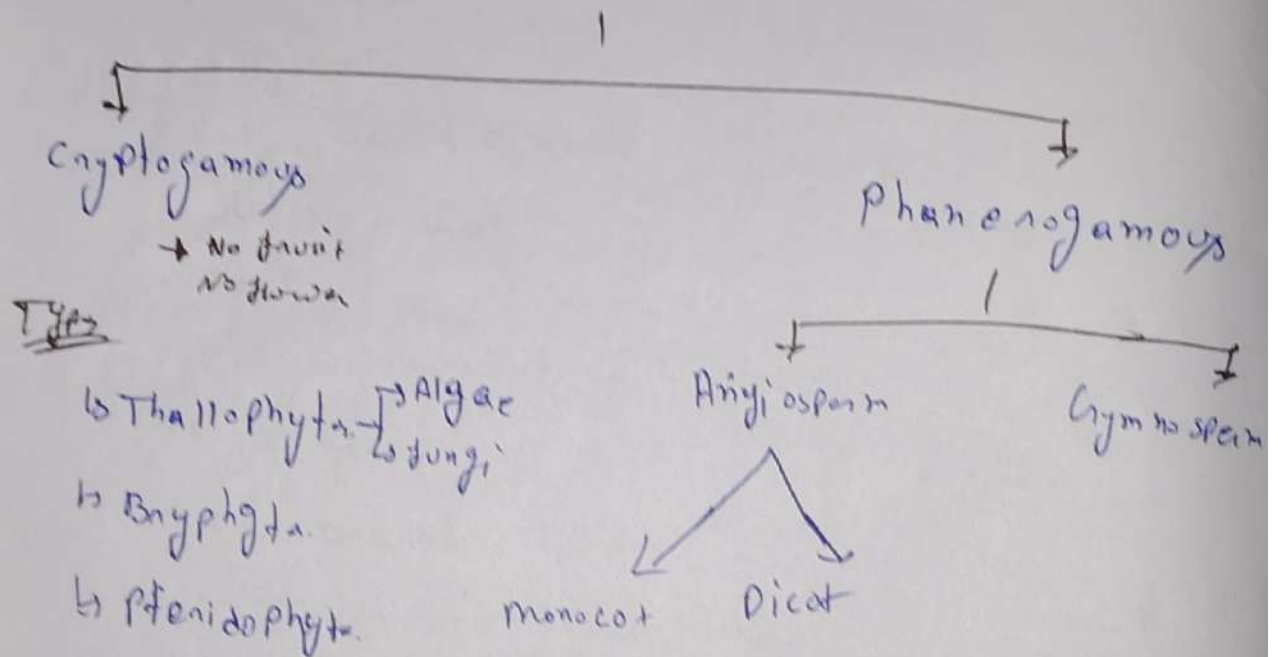


Botany

- Study of plants known as Botany
- Theophrastus father of botany
- Historia of Plantarum written by Theophrastus in which he describes 500 plants

Plant Kingdom



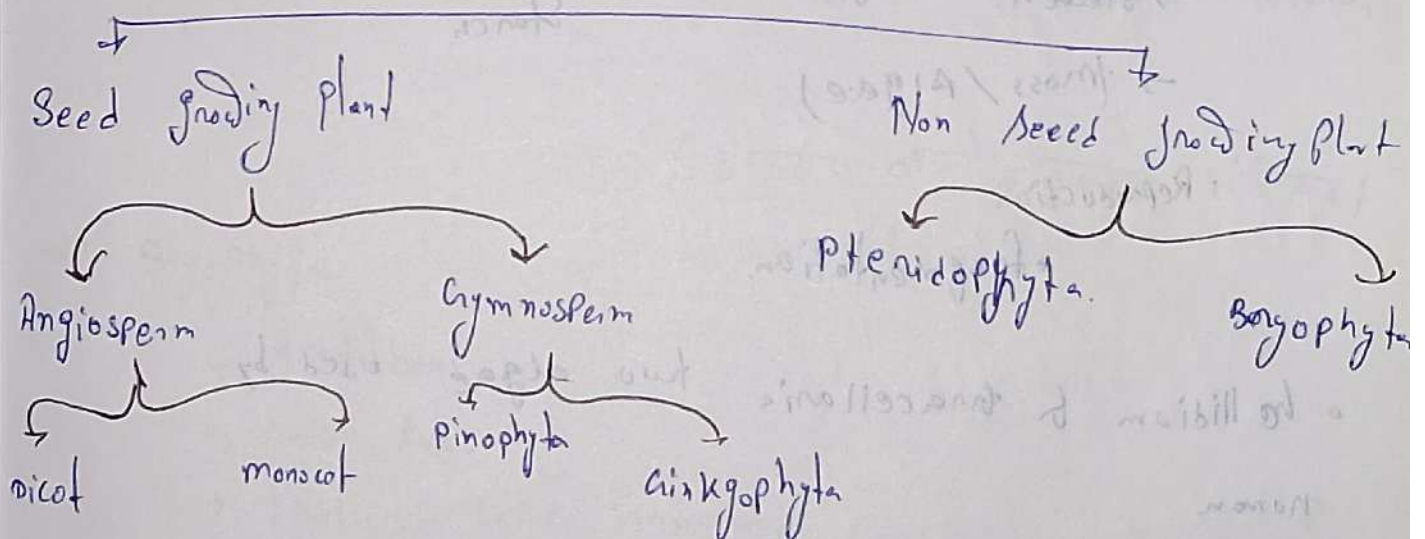
Cryptogamous

- Has No seeds
- Has No roots
No stems
on Leaves Ex - Algae
- Has some root and leaf-like Structure.
Ex - mosses and Liverworts
- Has roots stems and leaves
Ex - ferns.
- Horsetails

Phanerogamous

- Has seeds
 - Has no flower
↓
Needle like leaves
Ex - conifers
↓
wide leaves.
Ex - Ginkgo.
 - Has flower
Ex - Angiosperm

Plant



* Cryptogamous

↳ no seeds / no flower

- Thallus body

Thallophyta

↳ Thallus body

↳ seed. no

↳ flower no

↳ No vascular tissue

Xylem → Water

Phloem → food

Ex - Algae

* ALGAE

• study of Algae is known as phycology

• green have chlorophyll.

↳ cell wall ^{is made of} → cellulose

↳ ~~starch~~ store food in the form of starch

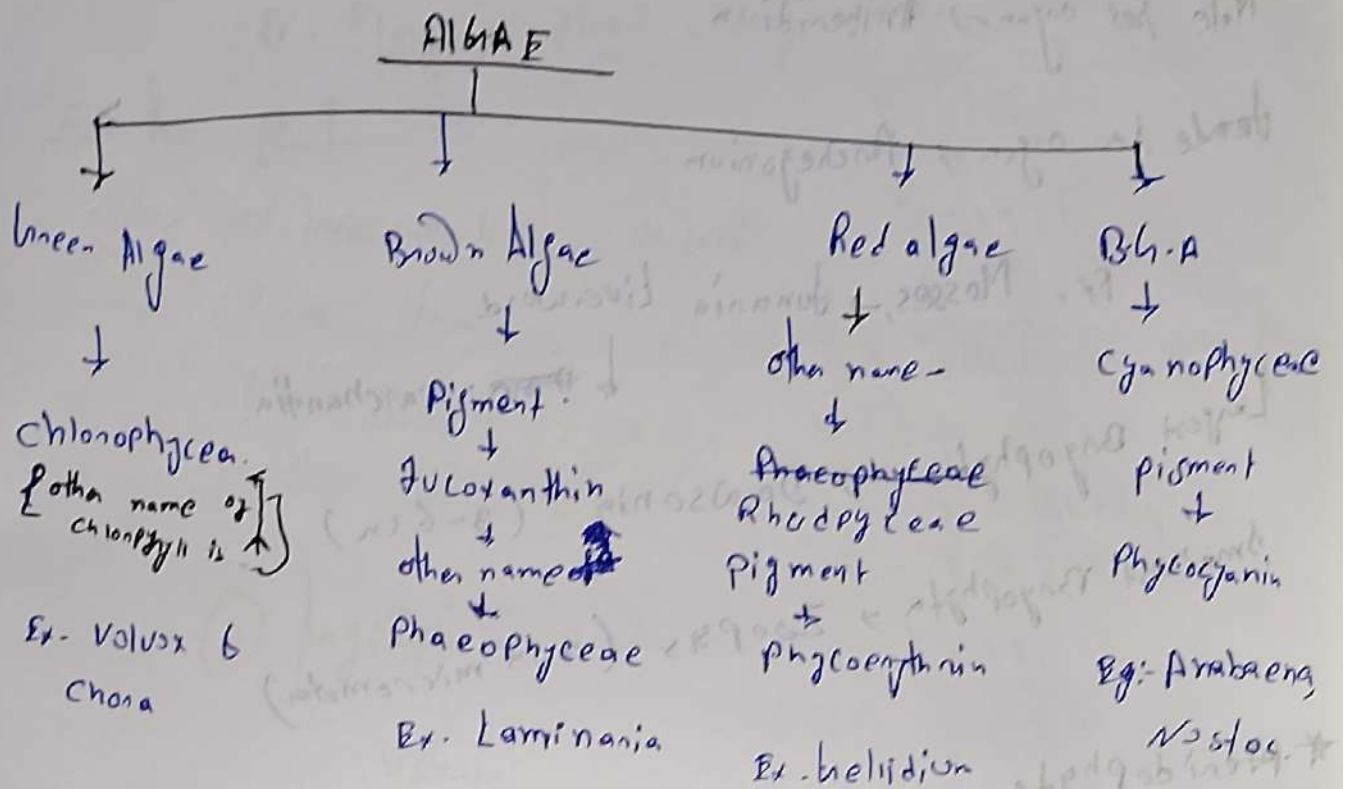
↳ (moss / Algae)

• Reproduction

• fragmentation

• Cellidium to Chlamydomonas two algae used by

human



* Lichen (Symbiotic relationship)

↳ Algae
↳ Fungi → Lichen.
↳ Natural Air Pollution indicator

(यदि कोई पौधा SO₂ के संपर्क में आएगा तो वह मर जाएगा)

* Bryophyta.

- Amphibian
- moist Area
- Thallus - Like structure
- No Vascular tissue
- Rhizoids Presence
- chlorophyll

Male sex organ $\rightarrow$ Antheridium

Female sex organ $\rightarrow$ Archegonium

Ex. Mosses, $\rightarrow$ Junonia, Liverwort

$\rightarrow$ ~~Marchantia~~ Marchantia

Largest Bryophyta $\rightarrow$ Dawsonia (3-6 in)

Smallest Bryophyta $\rightarrow$ Zoopsis (3-4 micrometers)

* Pteridophyta

$\hookrightarrow$ Land

$\hookrightarrow$ Xylem, Phloem

$\rightarrow$ Reptiles

$\hookrightarrow$ Ex - fern, ornament plant

* Phanerogamous.

② Gymnosperm -

$\hookrightarrow$ Gymno $\rightarrow$ Naked

$\rightarrow$ Gymnosperm $\rightarrow$ Gymnology

$\rightarrow$ Father of gymnosperm $\rightarrow$ Chamberlain

Ex - Strawberry, Pineapple.

Largest gymnosperm $\rightarrow$ Redwood or sequa
 $\hookrightarrow$ 400 feet.

Smallest $\rightarrow$ Zamia or Angion.

Ex. Eragroen plant, woody plant

In this plant.

→ True roots are present

→ True stem

→ True leaves

(ii) Angiosperm

↳ flower.

↳ covering

↳ Pollen grain + ovule

Largest Angiosperm → Eucalyptus → 100m

Smallest → Wolffia → (0.1 - 0.2 mm)

Monocot

Dicot

→ Single cotyledons.

→ Double cotyledons.

• एक बीजपत्र होता है,

• दो बीजपत्र होता है,

Ex - Grass, sugarcane,
Rice, corn

Ex - Mango, Pea, sunflower,
Gram, nut, Almonds,
Cashew, pulses

* Plant Hormone

↳ Total = 7

↳ main = 5

* Auxin :- Charles Darwin.

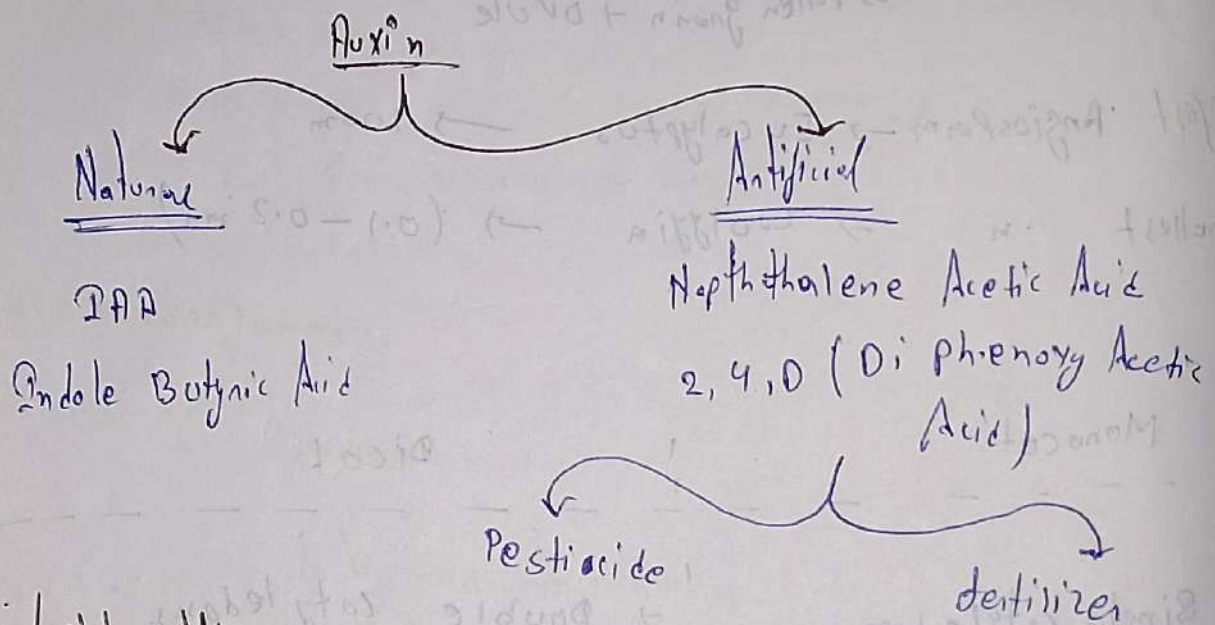
~~cell division~~

↳ cell division

↳ IAA (Indole-3 Acetic Acid)

↳ Growth hormone

↳ Apical buds



* gibberellin

• Kurosawa discovered gibberellin.

↳ GA3 main chemical

↳ Increase height and width in plants

↳ vascular tissue

↳ Increase stem strength

* cytokinin

- Discovered by Müller
- helps in cell division
- Auxin के साथ मिलकर कार्य करता है
- अचानक से गुणसूत्री में हुए बदलाव से रोकता है
(Prevent sudden changes in chromosomes)

* Ethylene

- gaseous form
- Fruit ripening
- respiration → Antidote is Acetylene

* Abscissic Acid

↳ Adicott

↳ Cotton plant Pinnule.

↳ termination ~~Pinna~~ ~~Pinna~~

~~है~~ , समय से पहले पतझड़ वाली होने देता है,

* florigens

- helps in blooming flowers.

* Transmuring →

- Damage part recover

Plant Disease

• Fungal disease

(i) Black root of wheat

(ii) Smell of Apple

(iii) Citrus disease

• Virus

T.M.V → Tobacco mosaic virus

P.M.V → Potato mosaic virus

Bunchy disease of Banana

Main of mosaic virus are potato में हवा
और P.M.V. and tobacco में हवा
time so on

• Bacteria disease:

• Tundu disease → wheat में होता है

• Red Root of corn

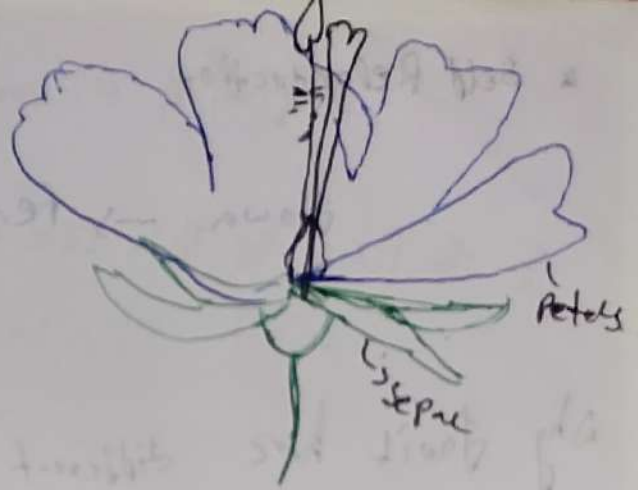
• Wilt disease → potato में होता है

• Citrus disease

* Plant morphology.

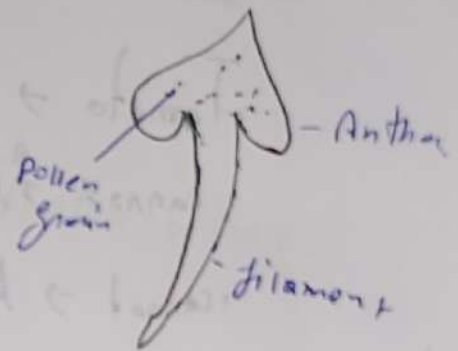
* flower.

Reproductive



• Androecium

- ✓ Anther
- ✓ Pollen grain
- ✓ Filament



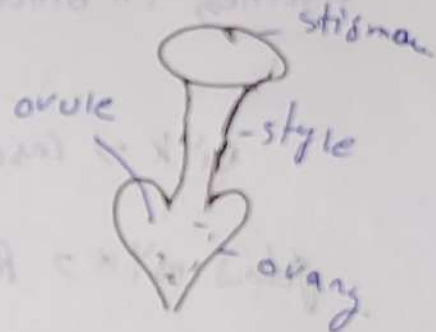
• male Reproductive part

• Gynoecium

✓ Stigma

- ✓ style
- ✓ ovule

✓ ovary



Female Reproductive part

Gynoecium

Reproduction in plants

• Pollination

Anther → pollen grain → Stigma

• Cross Pollination

Pollen grain → Stigma.

* Self Reproduction

flower $\rightarrow$ fertilization

Why Javit have different colour.

Tomato $\rightarrow$ Lycopene (Red)

cannot $\rightarrow$ Carotene (orange)

cannot $\rightarrow$ Anthocyanin (blue)

Why bitter found is Bitter —

Bitterness in Bitter found $\rightarrow$ momodericin

milk $\rightarrow$ casein (white colour)

yellow milk $\rightarrow$ Riboflavin & carotene

Green colour of potato $\rightarrow$ Solanine.

Cucumber $\rightarrow$ cucurbitacin

Green chilli $\rightarrow$ capsaicin

Red chilli $\rightarrow$ capsaicin

Papaya $\rightarrow$ Caricaxanthin (yellow colour)

Raddish $\rightarrow$ Iso cyanate

Berries
(~~Red & Blue~~) All have $\rightarrow$ Anthocyanine

Red & Purple colour अगर किसी भी सब्जी में हों
तो उसमें Anthocyanine पाया जाता है।

Turmeric $\rightarrow$ curcumin

Onion $\rightarrow$ Anthocyanine

oil $\rightarrow$ Allyl-Isothiocyanate (yellow colour)

Smell in onion $\rightarrow$ Sulfer

Beetroot $\rightarrow$ Betanin

Apple $\rightarrow$ Anthocyanin

Sunflower $\rightarrow$ xanthophyll